

Essay Outline and Guidelines

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Very Important Notice

1. Write the document of your essay according to your understanding.
Copying pasting from any source is not allowed!
2. You are allowed to use only Latex to write your essay.
3. Use AIMS Rwanda MSc Latex thesis template
4. Write your bibliography use a Latex .bib file.
5. Cite all sources of images, datasets and literature from which you took resources.

Content

1. Introduction
2. State-of-the-art
3. Problem Analysis
4. Methodology and Implementations
5. Case studies and Numerical Results
6. Summary, Conclusions and Future work
Bibliography

1. Introduction

1.1 Motivation

- give motivations for your essay work
- give relevance the field and describe a couple potential applications
- give a verbal description of your essay task and the problem you are addressing

- discuss some shortcomings of classical methods
- describe the methodology your used to address the problem in general terms

1.2 Objectives

- List a set of concise and achievable goals for your essay research within the given 6 week deadline
- List down your objective in the format "· to do this ...", "· to do that ..."

1.2 Thesis organization

Describe how your overall work is structured. This part could be written towards the end of your research work.

2. State-of-the-art

This chapter can have multiple chapters. It should contain literature reviews and the text to be written here refers to a broader context as it is are relevant to the topic of the essay.

- start with some historical accounts
- discuss chronological developments
- describe the state of the field in current time
- wherever necessary use mathematical presentations
- given comparative analysis of the various approaches in your own words
- discuss also future trend

3. Problem Analysis

- Describe each and every aspect of the specific problem your addressing, according to the title of your essay
- Give specific details
- Use mathematical descriptions
- Analyze the problem
- etc.

4. Methodology and Implementations

- describe and analyze the methods you are using to address you are problem
- substantiate why theses specific methods and approaches are most fitting to tackle your problem
- Present and discuss your (chosen) algorithms
- Explain your implementations, use a flowchart if possible, and provide code snippets

- if necessary
- etc.

5. Case studies and Numerical Results

- describe the concrete problem you solved
- the dataset used
- display the results of your implementation
- given analysis of the results
- explain challenges faced in solving the problems
- state the drawbacks of the methods used
- etc.

6. Summary, Conclusions and Future work

- Give the brief summary of your overall essay work
- Draw conclusions according the results you attained
- Explain what you could attained if given sufficient time, better resources, etc.
- Explain what future work will look like depending on the topic of your essay